

Markscheme

Mathematics: analysis and approaches
Practice question bank

76. (a)

A ball is dropped from a height of **4 m**. After each bounce it rebounds to **70%** of the height it fell from.

Find the total vertical distance travelled by the ball by the time it hits the ground for the 6th time, correct to 3 significant figures.

the distance is the initial fall plus twice each bounce height up to the 6th impact: (M1)

$$4 + 2(4(0.7) + 4(0.7)^2 + 4(0.7)^3 + 4(0.7)^4 + 4(0.7)^5) \quad \text{span style="float: right;">(M1) A1}$$

evaluate the geometric sum inside the brackets (M1)

$$\approx 19.5 \text{ m} \quad \text{span style="float: right;">A1}$$

[6 marks]